

**APSC 2113 - Engineering Analysis I**  
**Instructor: Professor Kartik Bulusu, MAE Dept., GWU**  
**LQ4 - Fall 2026**

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Name: \_\_\_\_\_

Student ID: \_\_\_\_\_

**Instructions**

- Answer all questions. Select **one** answer for each question. Simplify all answers completely.
  - The duration of the test is 7 minutes. No calculators, notes, or other electronic aids unless explicitly permitted.
  - Write your name and student ID clearly at the top of the page. You are allowed to collaborate with two students of your choice during this LQ.
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**Questions**

**Q1.** The characteristic roots of an eighth-order ODE are given by

$$-5, -4, -3, -2, -1, 0, 1, 2.$$

What is the general solution of the ODE with time  $t$  as the independent variable?

For a linear constant-coefficient ODE, each real characteristic root  $r$  produces a solution of the form

$$y_r(t) = Ce^{rt}.$$

The characteristic roots are

$$r = -5, -4, -3, -2, -1, 0, 1, 2.$$

Therefore, the corresponding independent solutions are

$$e^{-5t}, \quad e^{-4t}, \quad e^{-3t}, \quad e^{-2t}, \quad e^{-t}, \quad 1, \quad e^t, \quad e^{2t}.$$

Since all eight roots are distinct, the general solution is the linear combination of these eight solutions:

$$y(t) = C_1e^{-5t} + C_2e^{-4t} + C_3e^{-3t} + C_4e^{-2t} + C_5e^{-t} + C_6 + C_7e^t + C_8e^{2t}$$